🌿 Algae & Photosynthesis: Reading · ANSWER KEY

Biology A · Unit 2 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A1 algae + microscope, A2 algae lab, A4 Pop beads, Elodea & snails, A4.5 Stomata

Name: _____

How Algae Turn Light into Food

Algae are plant-like living things that live in water. Some are big, like kelp; most are tiny, like phytoplankton and pond scum. To see tiny algae, we use a microscope. Total magnification is the eyepiece (10x) times the objective lens. A 40x objective gives 40 × 10 = 400x. Always find the object on low power first, and clean lenses only with lens paper.

Algae and plants make their own food through photosynthesis. They take in carbon dioxide and water and use light energy to make glucose, a sugar that stores chemical energy. Oxygen is released. The word equation is: carbon dioxide + water + light → glucose + oxygen.

Photosynthesis happens in the chloroplasts, which contain green chlorophyll. It has two stages. In the light reactions (in the thylakoids), light energy splits water and oxygen is released. In the Calvin cycle (in the stroma), carbon dioxide is built into glucose.

Leaves have tiny pores called stomata, mostly on the underside. Guard cells open and close the stomata. Carbon dioxide comes in, and oxygen and water vapor go out. In hot, dry weather the stomata close to save water.

BTB is a color indicator for carbon dioxide. It turns yellow when CO2 goes up and blue when CO2 goes down. Snails make CO2 by respiration; Elodea in the light uses CO2 by photosynthesis. In the algae lab, fertilizer helped algae grow, but dish soap did not.

Check your reading

1. Write the word equation for photosynthesis.
Carbon dioxide + water + light → glucose + oxygen.

2. What do guard cells do?
They open and close the stomata.

3. A tube of BTB turns yellow. What does that tell you?
Carbon dioxide went up.